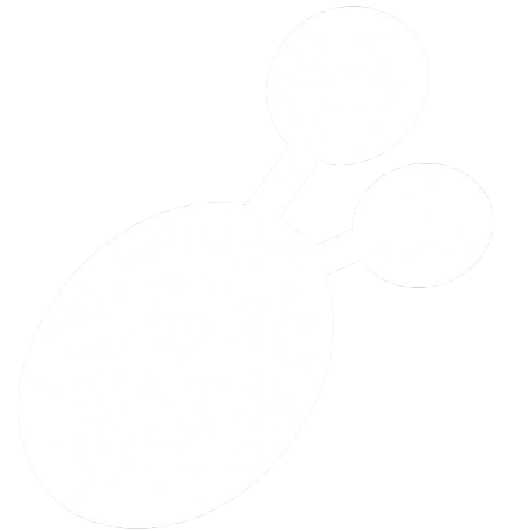


Intro to the Cluster

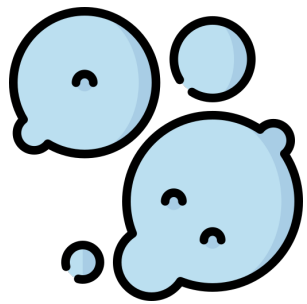
LaBella Lab
Wild Yeast Project

Outline

- Brief summary of ONT (i.e., Nanopore)
- Intro to Practice Activity
- Practice Activity

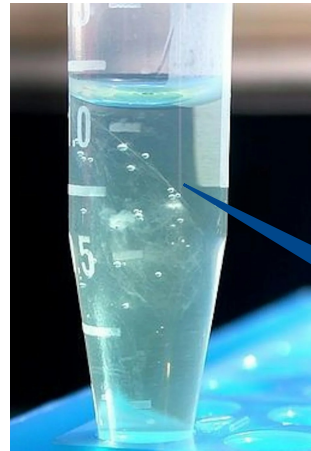


DNA Sequencing



Yeast

DNA
Extraction
→



Purified DNA =
Many copies of the
same sequence(s)

Primate mtDNA

Project of "Primate mtDNA"Character Matrix "Character Matrix"

GraphicsTextParametersModulesCitationsSearch Data

Project

Primate r

File Incorpor

New...

Taxa

Charac

Trees f

Character

Taxon \ Character

1 Homo sapiens

2 Pan

3 Gorilla

4 Pongo

5 Hylobates

6 Macaca fuscata

7 M. mulatta

8 M. fascicularis

9 M. sylvanus

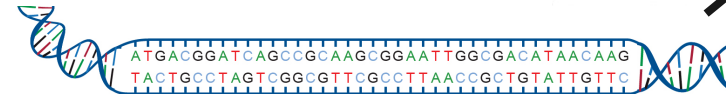
10 Saimiri sciureus

11 Tarsius syrichta

12 Lemur catta

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

A A



DNA
Sequencing
↗

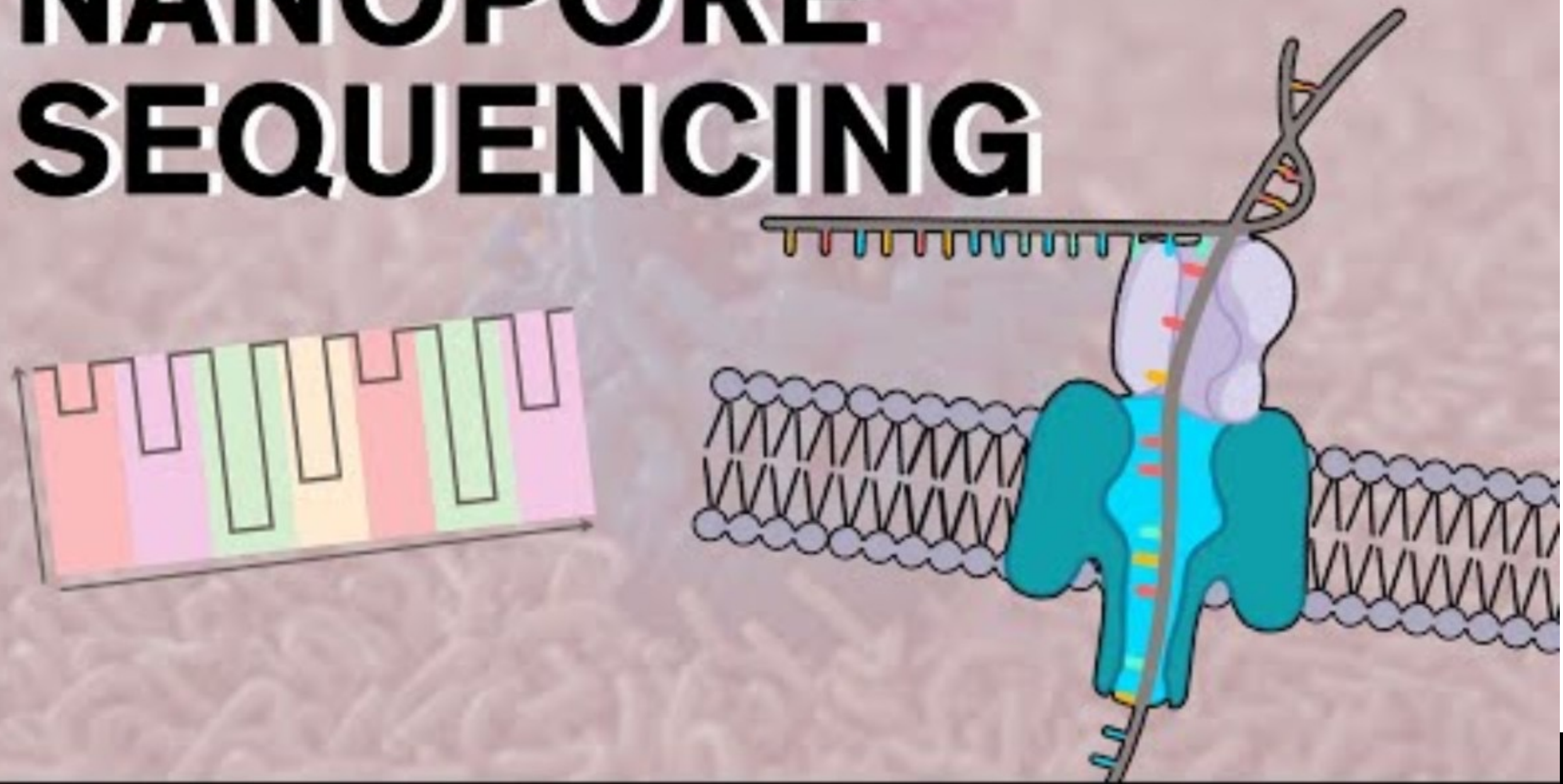


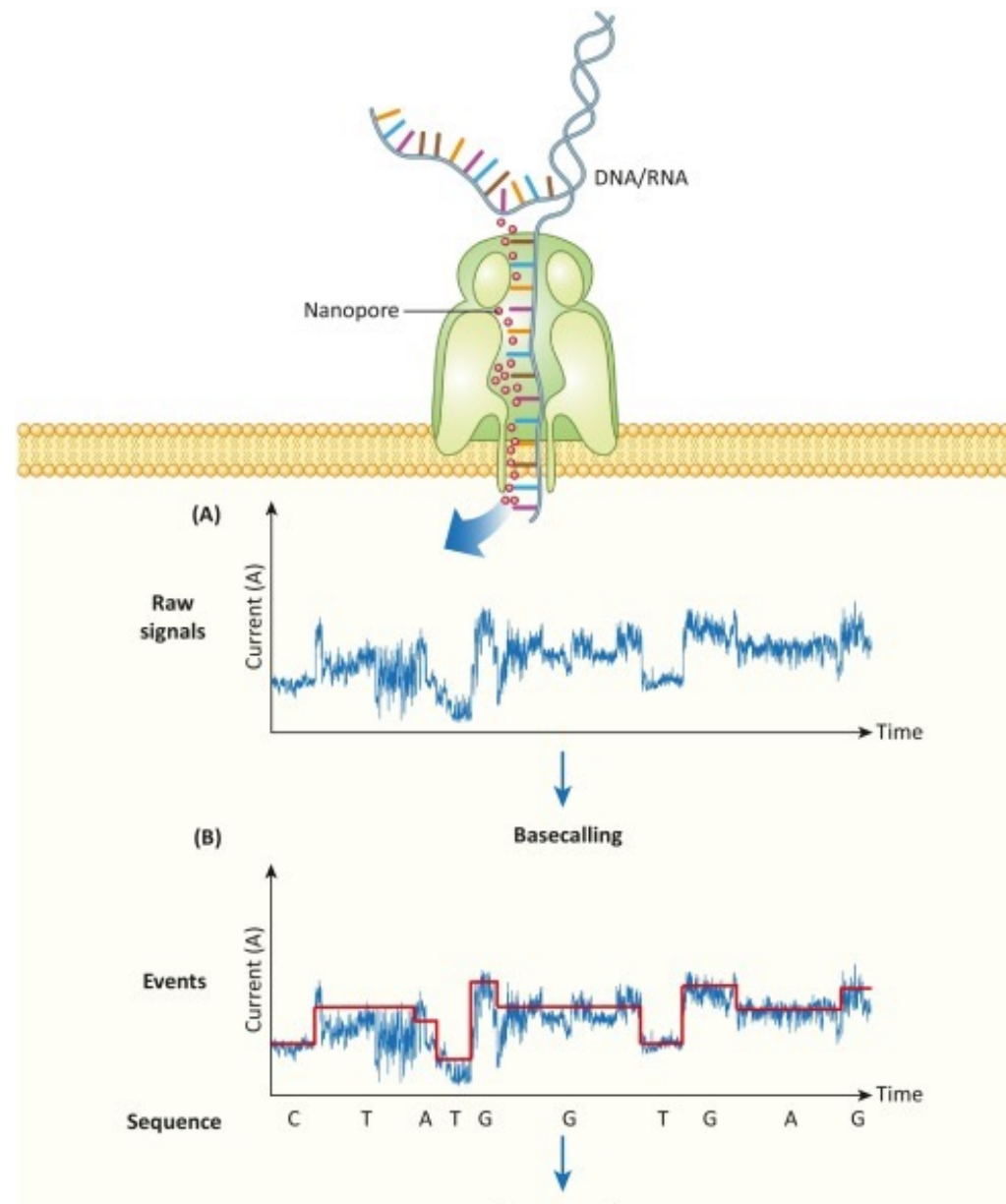
Oxford Nanopore Sequencing

- Completely novel technology that does not require polymerase
- Does not sequence by synthesizing a new strand of DNA
- Holds the world record for longest DNA sequence (2.3 M bp)
- High (but not highest) accuracy



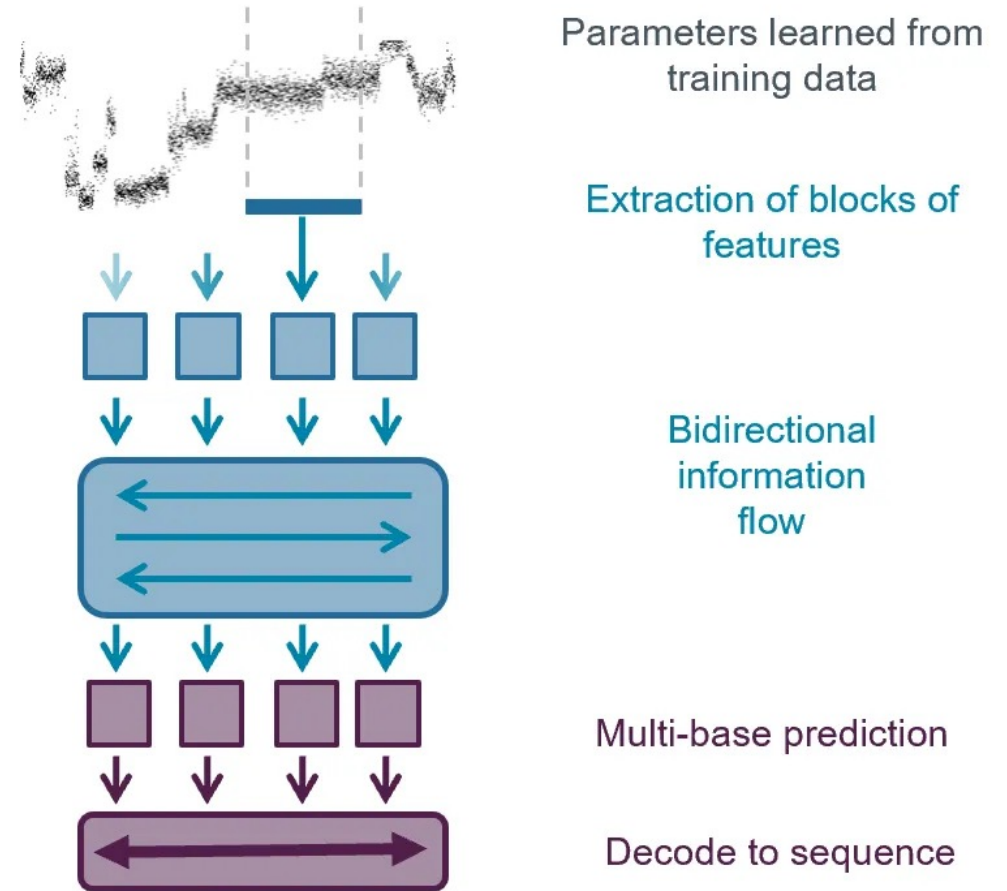
NANOPORE SEQUENCING



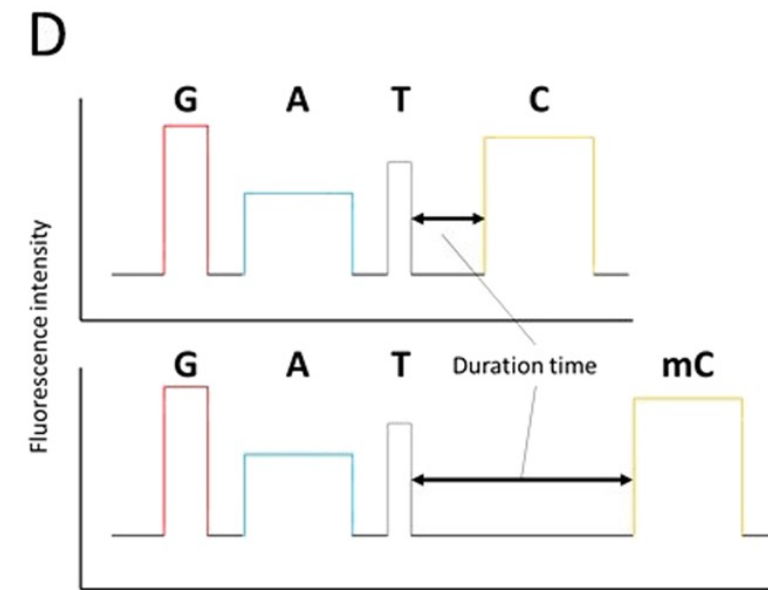
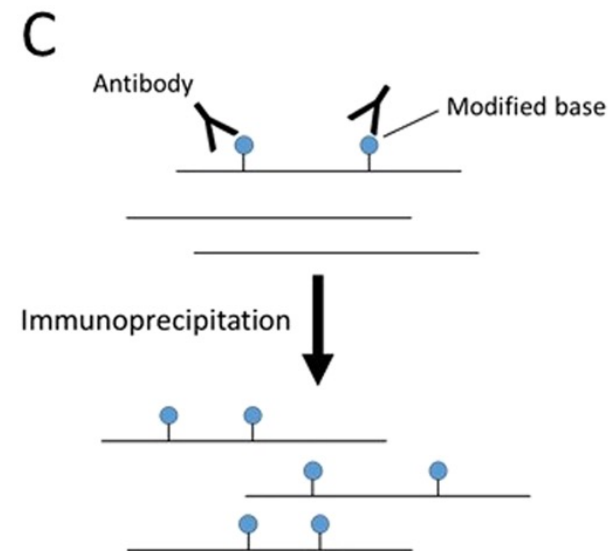
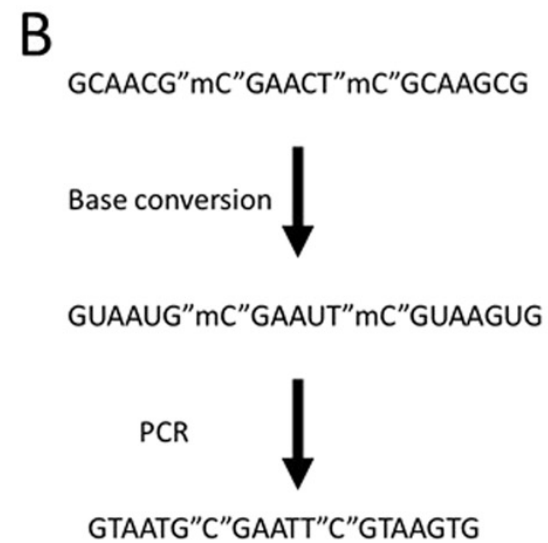
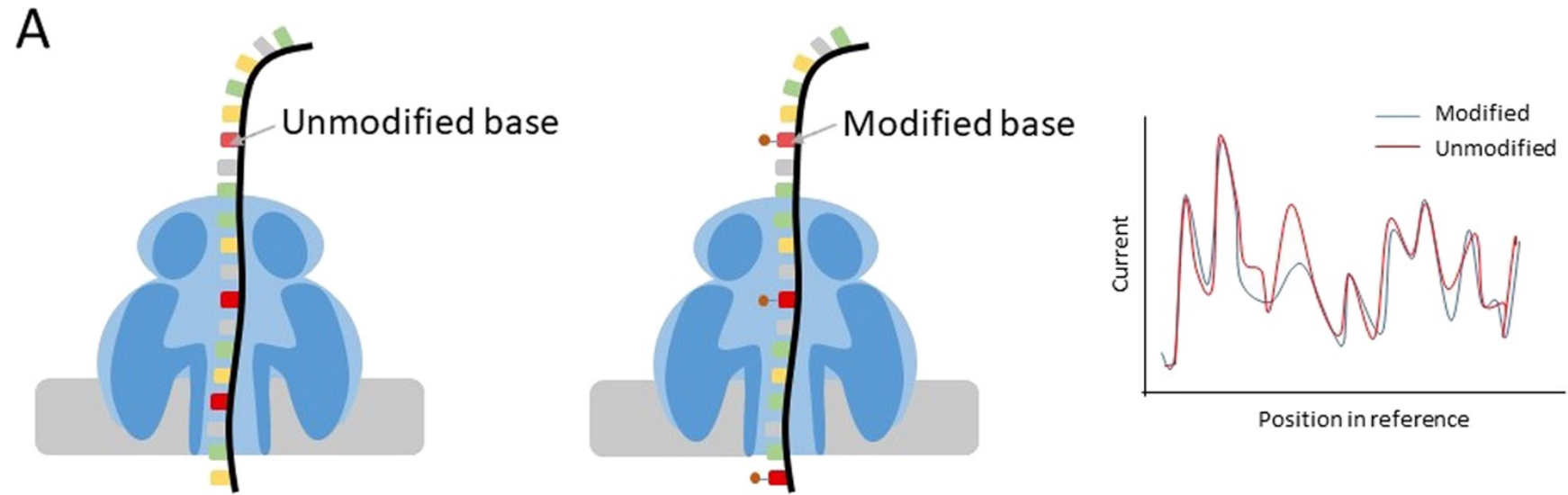


Base Calling – Neural Networks

- Machine learning model trained on known data
- New models are constantly being developed



Detects modified bases



Practice Activity

https://github.com/The-Lab-LaBella/Wild_Yeast_Summer_2026/blob/main/Intro_to_Cluster.md

